

Internalizing Environmental Risk: Insurance Design and Firm Behavior in Hazardous Industries

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Motivation

- ▶ USTs sit beneath $\sim 500,000$ U.S. facilities and are the leading source of groundwater contamination (EPA, 2024); mean cleanup $\sim \$420,000$
- ▶ Strict liability internalizes damages *only* when $P_i \propto \text{Risk}_i$ (Shavell, 1982). **38 states charge a flat fee $\bar{P}$:** zero marginal cost of aging
- ▶ Upgrading wall construction is irreversible: SW $\rightarrow$ DW requires full tank replacement ($\sim 60\text{--}90\%$ premium; GAO, 1987)

Question. Does replacing flat-premium public insurance with risk-based private coverage induce firms to internalize environmental damages?

Today's Punchline

Reduced form (*alive-at-reform sample, $N \approx 9.8M$ tank-years*)

- ▶ TX $\times$ Post raises annual tank-closure probability by +1.58 **pp** on a 2.0% baseline
- ▶ 4 $\times$ larger effect in pre-1989 vintage (+3.1 pp) — the high-risk SW-heavy stock responds most
- ▶ LUST discovery falls -1.27 **pp** on a 1.73% baseline; both background and inspection-triggered drop
- ▶ Conditional on closing: +53% relative shift to the **replacement** margin (12.0% $\rightarrow$ 18.3%); aggregate exit volume unchanged

Structural (4-parameter DDC)

- ▶ $\hat{\kappa} = \$261k$, $\hat{K} = \$52k$, $\hat{\gamma}_p = -0.29$, $\hat{\gamma}_r = 0.16$
- ▶ All three counterfactuals (Subsidy / Pigouvian / Mandate at age 25+) reduce social welfare at $E = \$17k$ (Marcus, 2021); optimal K subsidy $s^* \approx 10\%$, far below 50%

What I Do

Reduced form. Difference-in-differences exploiting the 1999 TX insurance reform on a 17-state CEM-matched panel; facility FE + cell $\times$ year FE absorb the 1998 EPA-deadline spike *within* tank-vintage cells. Active-at-treatment sample restriction.

Structural. Dynamic discrete-choice model: maintain / exit / replace. Recovers $(\kappa, K, \gamma_p, \gamma_r)$ via Nested Pseudo-Likelihood (Aguirregabiria & Mira, 2007).

Counterfactuals. K subsidy, Pigouvian surcharge $h(a) \cdot E$, and an age-based mandated-closure rule. Welfare = firm surplus – external damage PV – govt outlay PV.

A Dynamic Model of Tank Closure

A Dynamic Decision: Maintain or Close

Each period t , the facility chooses $d_t \in \{\text{Maintain}, \text{Close}\}$.

Flow utility (Maintain): $u(a) = R - C_j(a)$ — revenue net of insurance premium under regime j .

Bellman:

$$V(a) = \max \left\{ \underbrace{u(a) + \beta \mathbb{E}[V(a') \mid a]}_{\text{continuation}}, \underbrace{\kappa}_{\text{scrap}} \right\}$$

Stopping rule: close if $V_{\text{cont}}(a) < \kappa$.

Assumption — full coverage: environmental risk enters as an *ex-ante* premium cost, not an *ex-post* shock.

Three Regimes, Three Exit Ages

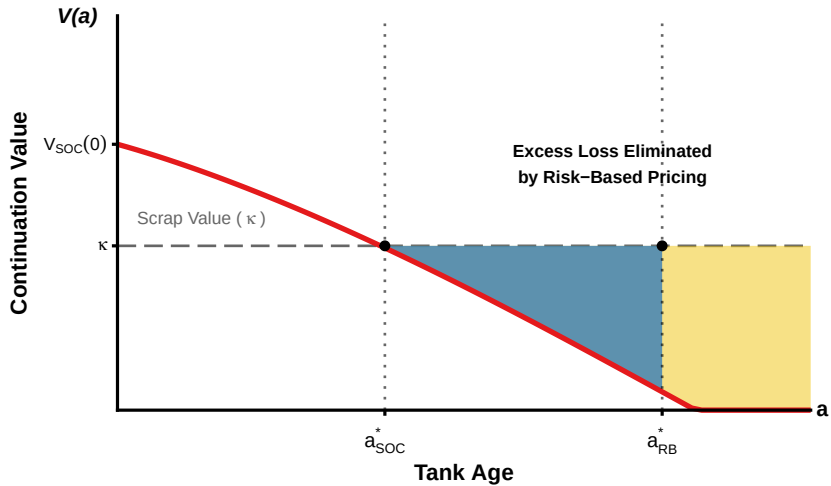
Flat-fee pooling (FF) — 38 states. ; $C(a) = \bar{P}$, so $\partial C / \partial a = 0$. No private incentive to close as the tank ages.

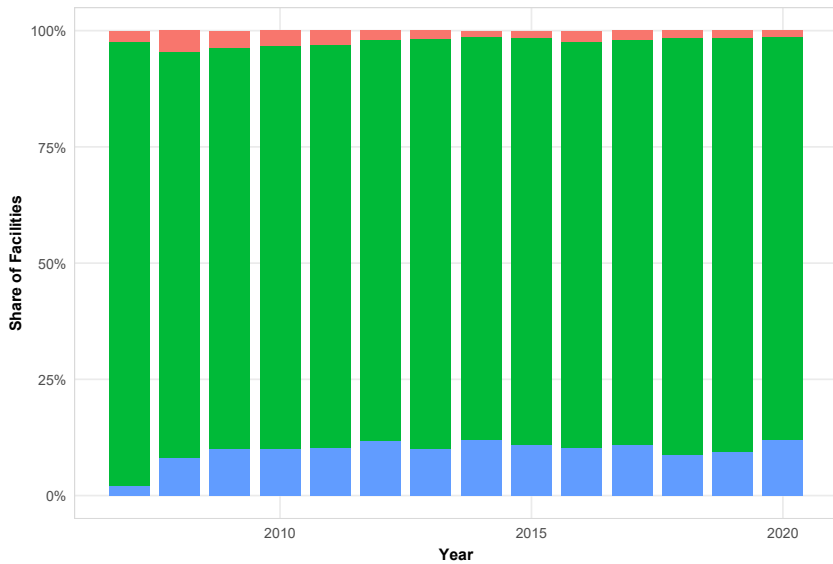
Risk-based pricing (RB) — Texas post-1999. ; $C(a) = \lambda \cdot h(a) \cdot L$, so $\partial C / \partial a > 0$. Premium rises with the leak hazard.

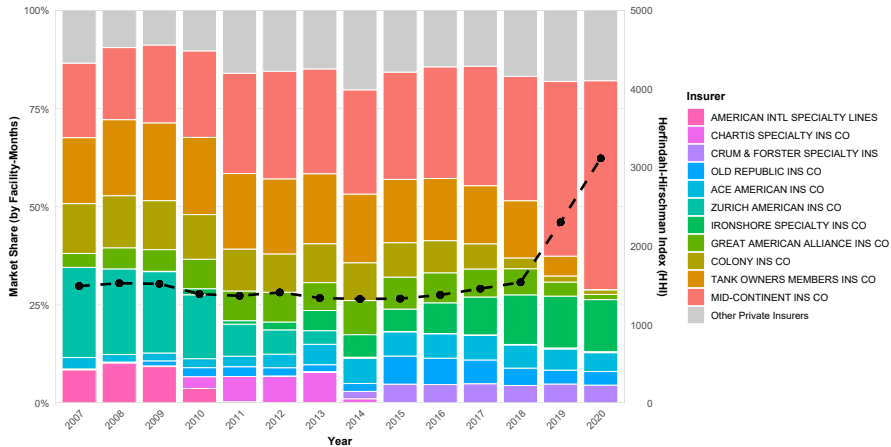
Social optimum (SOC). ; $u_{\text{SOC}}(a) = R - h(a) \cdot (L + E)$. Planner internalizes both remediation L and unpriced health damages E (contamination is unobservable; Marcus 2021).

$$a_{\text{SOC}}^* < a_{\text{RB}}^* < a_{\text{FF}}^*$$

RBP closes the gap *partially*. The residual wedge $a_{\text{RB}}^* - a_{\text{SOC}}^*$ is driven by the unpriced externality E .







Premium Construction

Texas — rebuilt from Mid-Continent Casualty rate filings (SERFF, 2006–2024):

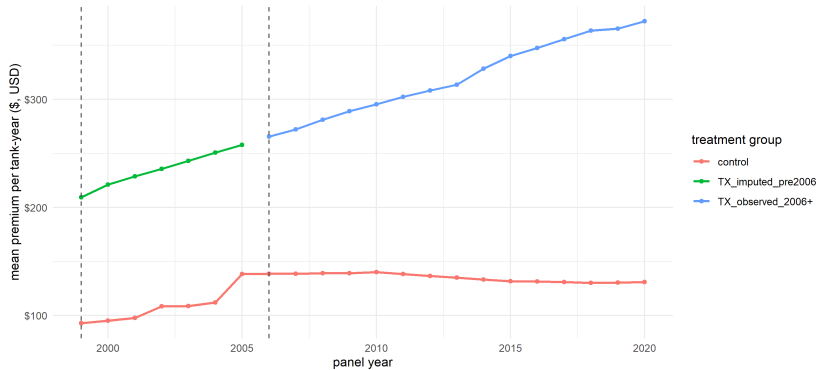
$$P_{ijt} = \text{Base}_t \times \text{ILF}_{it} \times \left(1 + \sum_k \lambda_{k,t} X_{ijt,k}\right)$$

- ▶ Base_t : per-tank base rate, four filing periods
- ▶ ILF_{it} : increased-limit factor; fix coverage at federal-minimum \$1M/\$1M
- ▶ X_{ijt} : tank age, wall, piping, leak-detection technology
- ▶ $\lambda_{k,t}$: additive credits/debits from the filed schedule

Control states — state financial-responsibility fund records: a per-tank annual assessment, largely flat across years and tank characteristics.

Mean per-tank premium by year x treatment group — RAW DATA

TX_imputed_pre2006 = engine-imputed Mid-Continent premium for 1999-2005



Risk Prediction: Setup

Estimand. Annual first-release hazard in the never-yet-leaked risk set:

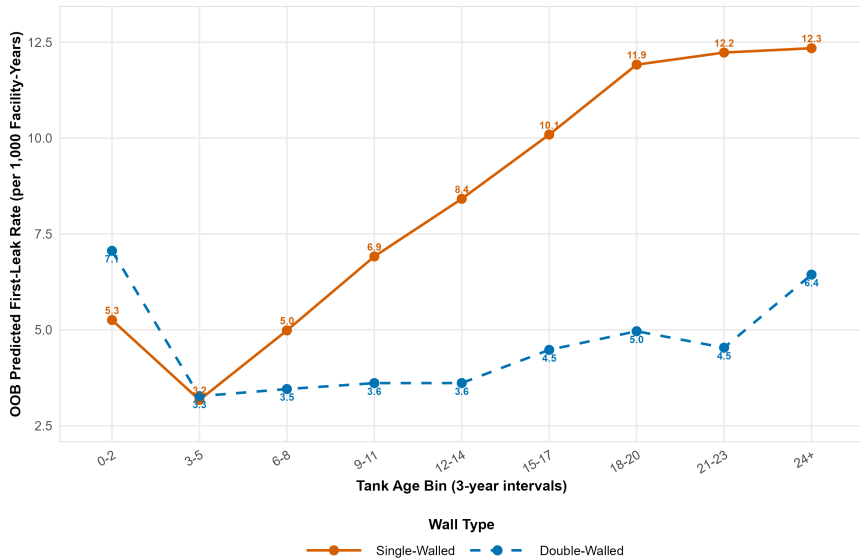
$$h(t \mid x) = P(\text{first confirmed release in year } t \mid \text{no prior release, } x)$$

In plain words: given what an underwriter can observe at the start of a policy year, what's the probability of a claim?

Model. Penalized logistic regression on the pre-treatment panel:

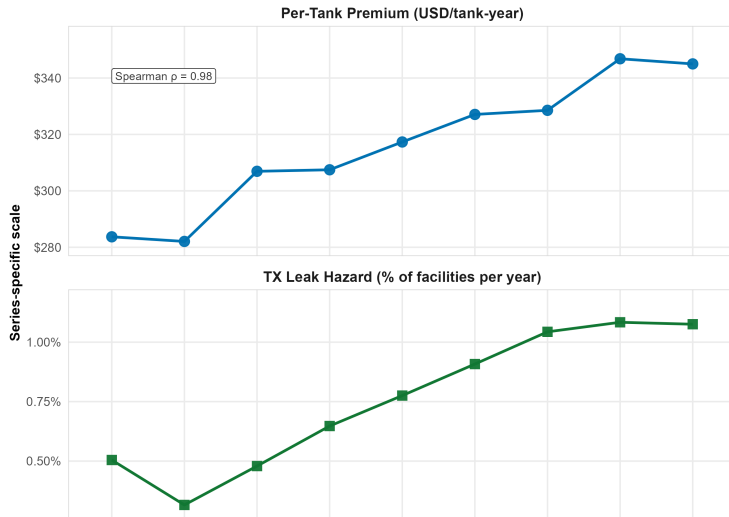
- ▶ x : 3-year age bins fully interacted with wall construction; state FE; fuel and portfolio controls
- ▶ Lasso variable selection; inverse-frequency weights for the 0.70% base rate
- ▶ Out-of-sample CV (10 folds); post-hoc recalibration to the empirical event rate

Sample. 3.93M pre-treatment facility-years, 27{,}455 confirmed first-release events. $\hat{h}(x)$ feeds the next figure and the structural-model hazard $h(a_t, W_i)$.

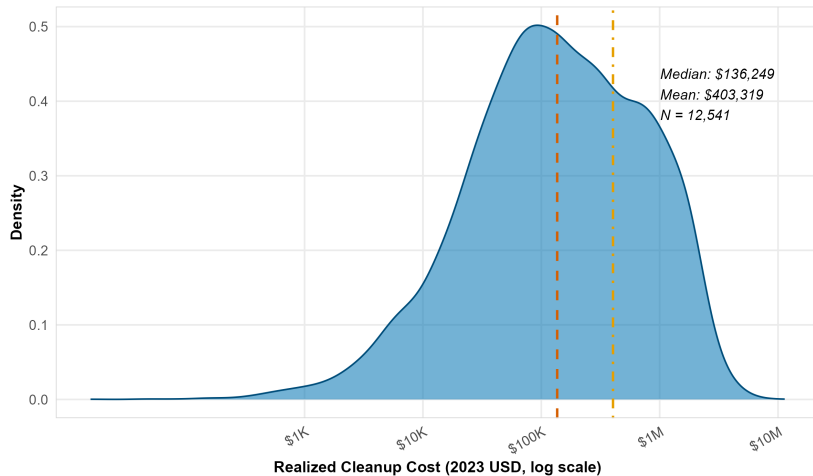


Premium Schedule Tracks Texas-Estimated Leak Hazard

Per-tank premium (TX Mid-Continent, 2006-2014) and TX leak hazard (01n elastic net OOB) both rise across age bir



Pooled Cleanup Cost Distribution



	Texas	17 control states
<i>Panel size</i>		
Facility-years (incumbent, 1985–2018)	870,341	3,782,424
Tank-years (active-at-treatment)		9,761,982
<i>Tank stock at Dec 22, 1998 (per facility)</i>		
Facilities	25,985	114,304
Mean tanks	2.78	2.74
Single-walled share (tank-weighted)	89.1%	89.5%
Mean oldest tank age, years	15.32	18.09
Mean total capacity, gal	22,585	62,405
Gasoline facility share	66.1%	75.4%
<i>Vintage composition (facility counts, full panel)</i>		
Pre-1989 vintage (faced EPA phase-in)		96,923
Post-1988 vintage		42,523
<i>Pre-reform annual rates (control mean)</i>		

Sample and Specification

Sample. Tank-year panel; alive-at-reform restriction $A_f = 1$ (facility had ≥ 1 open tank on Dec 22, 1998). 18 study states (TX + 17 control); CEM-matched on portfolio at reform. $N \approx 9.8\text{M}$ tank-years.

Static DiD.

$$Y_{it} = \beta (\text{Texas}_{f(i)} \times \text{Post}_t) + \alpha_i + \delta_{ct} + \varepsilon_{it}$$

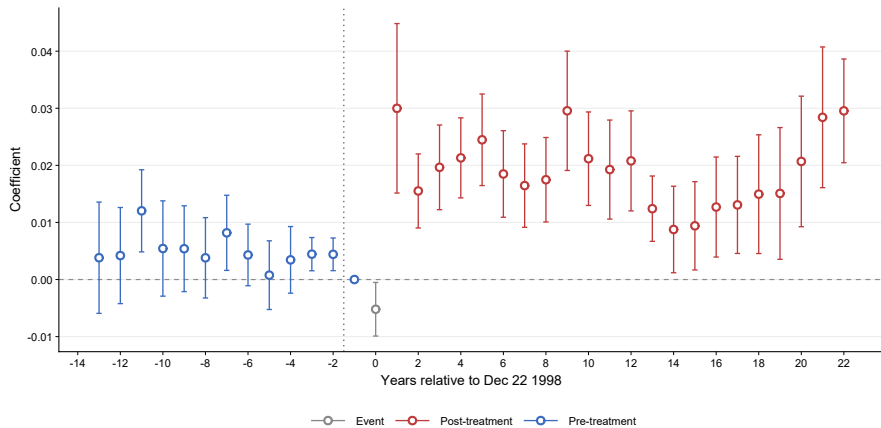
Event study (replace Post_t with the full set of $1[t - 1998 = \tau]$ interactions, $\tau \neq -1$).

- ▶ α_i : facility FE. δ_{ct} : cell $\times$ year FE; $c = \{\text{wall} \times \text{fuel} \times \text{capacity} \times \text{install cohort}\}$
- ▶ β is the ATT on annual closure probability, averaged over **Texas tanks at facilities active on Dec 22, 1998**
- ▶ *Identifying assumption*: conditional parallel trends within the cell $\times$ year FE. SEs clustered by state.

ATT on Tank Closure: +1.58 pp on a 2.0% Baseline

	(1) Raw	(2) + Fac FE	(3) + Cell FE
Constant	0.0203*** (0.0022)		
did_term	0.0106*** (0.0022)	0.0338*** (1×10^{-6})	0.0158*** (0.0040)
Observations	9,761,982	9,761,982	9,761,044
R ²	0.00047	0.03550	0.07745
Within R ²		0.00218	0.00036
panel_id fixed effects		✓	✓
cell×year FE			✓

Cols add facility FE, then cell×year FE. Col (3) is the headline.



Heterogeneity: Pre-89 Vintage Drives the Effect

	(1) Raw	(2) + Fac FE	(3) + Cell FE
Constant	0.0103*** (0.0011)		
did_term	0.0048*** (0.0011)	0.0117*** (0.0028)	-0.0067** (0.0031)
pre89_cohort	0.0135*** (0.0023)	0.0671*** (0.0062)	
did_term $\times$ pre89_cohort	0.0094*** (0.0023)	0.0382*** (0.0029)	0.0309*** (0.0065)
Observations	9,761,982	9,761,982	9,761,044
R ²	0.00249	0.04685	0.07772
Within R ²		0.01392	0.00065
panel_id fixed effects		/	/

LUST Discovery: Both Channels Fall

Outcome (incumbent sample)	Pre-reform ctrl mean	TX $\times$ Post
Total leak discovery (0/1)	1.73%	−1.27 pp ^{***} (0.0032)
<i>Decomposition (Primary 0–60d window):</i>		
Background	1.16%	−0.76 pp ^{***} (0.0021)
Inspection-triggered	0.61%	−0.51 pp ^{**} (0.0017)
<i>N</i> facility-years	4,650,240	
Sample	Incumbent (alive at Dec 22, 1998)	
FE	Facility + make_model_fac $\times$ year; cluster by state	

Caveat: LUST event-study pre-trend not flat (TX leaks converging down pre-1998 due to mandate compliance). Static coefficient partly reflects that trend; see appendix ES.

What Closing Firms Do: Composition Shifts to Replacement

Outcome (conditional on closing)	Pre-reform ctrl mean	TX $\times$ Post
Facility closes all tanks this year	70.9%	+6.5 pp ^{***} (0.019)
Has a permanent closure (no replace)	88.3%	-6.2 pp ^{***} (0.012)
Has a replacement closure (upgrade)	12.0%	+6.3 pp ^{***} (0.013)
Any DW tank installed	4.1%	-1.4 pp [*] (0.007)
Net tank change (count)	-1.905	+0.020 (0.085)
Capacity change (gal)	-100,394	-1,397 (5,715)
<i>N</i> closing facility-years	59,107	
Sample	Incumbent, any_closure = 1	
FE	Facility + make_model_fac $\times$ year; cluster by state	

Structural Model

Model Setup

State. $s_t = (a_t, W_t, R_t, \varepsilon_t)$ — 9 age bins $\times$ 2 wall types $\times$ 2 regimes = 36 states. Actions: maintain / exit / replace.

Flow utility (numeraire $R = \$10\text{k/yr}$).

$$u^m(s) = 1 - \gamma_p p_t - \gamma_r h(a_t) L$$

$$u^e(s) = \kappa \quad (\text{scrap value})$$

$$u^r(s) = -K \quad (\text{install new DW tank})$$

Calibrated upstream (*outside the model*): premium p_t , hazard $h(a_t)$, liability L — all from the descriptive-evidence section.

Estimated jointly (*inside the model*): $\theta = (\kappa, K, \gamma_p, \gamma_r)$. NPL match of choice shares across (a, W, R) cells.

Identification

Each parameter is identified by a distinct moment in the choice data:

- ▶ κ : the unconditional exit rate $P(\text{exit} \mid s)$
- ▶ K : the replace-vs-exit share among closing tanks
- ▶ γ_p : cross-cell response of choice shares to premium variation in (a, W, R)
- ▶ γ_r : divergence in age-closure hazard between FF and RB regimes

NPL (Aguirregabiria & Mira 2007); AM-2002 profile-likelihood SEs.

Parameter Estimates

	Observed TX 2006+, controls 1999+	Extended (2000+) TX 2000+, controls 2000+
κ_{exit} (\$)	\$261,222 (\$306)	\$257,702 (\$310)
K (\$)	\$51,779 (\$142)	\$52,658 (\$146)
γ_{price}	-0.293 (0.049)	-1.142 (0.048)
γ_{risk}	0.164 (0.003)	0.153 (0.003)
$\log L$	-268,481	-247,614
N obs	2,282,735	2,300,780

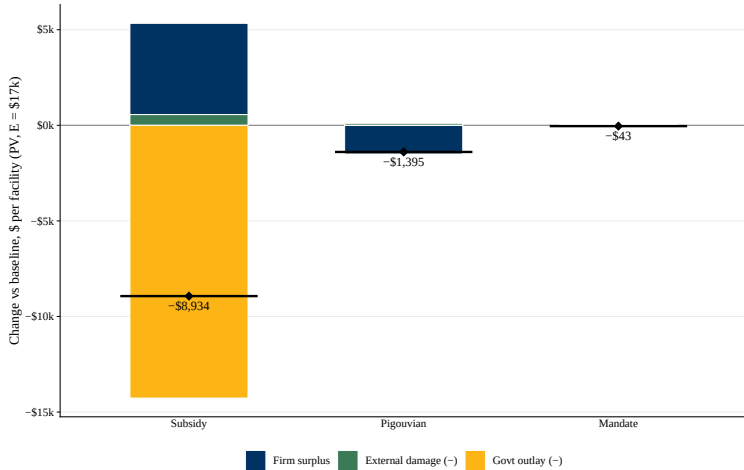
SEs are AM (2002) profile-likelihood; bootstrap SEs deferred for a future pass.

$\hat{\kappa} = \$261k \approx 26 \times \text{annual revenue (land-value liquidation range)}$. $\hat{K} = \$52k \approx 5 \times \text{revenue (tank-replacement cost)}$. $\hat{\gamma}_{\text{price}} = -0.29$: 1% of revenue in extra premium $\approx 0.3\%$

Counterfactual Policy Scenarios

Policy	Mechanism
Subsidy	$K \rightarrow 0.5 K$ — replacement cost halved
Pigouvian	Add $h(a) \cdot E$ to maintain cost; $E = \$17,000$ (Marcus, 2021)
Mandate	Force $P(\text{maintain}) = 0$ for $a \geq 6$ (age 25+)

Procedure. Re-solve Bellman at θ^{CF} via NPL; compute QSD; integrate welfare = firm surplus — external damage PV — govt outlay PV.



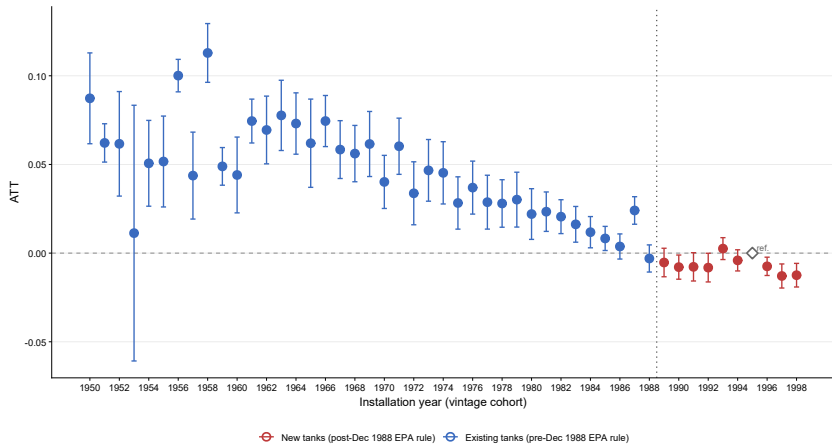
Counterfactual Welfare Summary

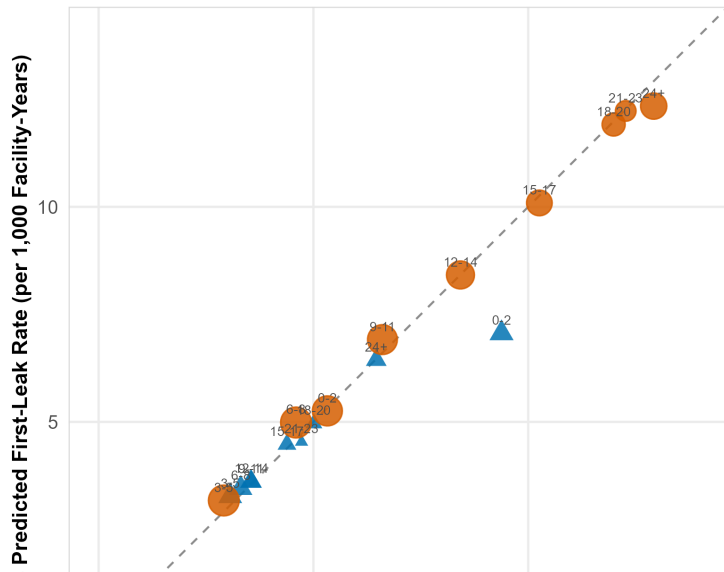
	$\bar{P}(M)$	$\bar{P}(E)$	$\bar{P}(R)$	Firm Surplus	Social Welfare	ΔW
Baseline	98.0%	1.7%	0.2%	\$307,724	\$305,344	$\pm \$0$
Subsidy	96.2%	1.1%	2.8%	\$312,498	\$296,409	-\$8,934
Pigouvian	97.7%	2.0%	0.2%	\$306,225	\$303,948	-\$1,395
Mandate	98.0%	1.8%	0.2%	\$307,633	\$305,301	-\$43

All three CFs reduce social welfare at $E = \$17k$. Subsidy hurts most: govt outlay outweighs firm-surplus gain. Mandate roughly neutral.

Takeaways

1. **Risk-based pricing does shift behavior.** A +1.58 pp annual closure ATT — about $4\times$ larger in the pre-1989 SW-heavy stock that the schedule actually prices.
2. **The shift is on the composition margin, not exit volume.** Conditional on closing, the **replacement** rate nearly doubles ($12\% \rightarrow 18\%$); net tank counts and capacity barely move. LUST discovery falls through both background and inspection-triggered channels.
3. **Welfare under the priced regime is still below first-best.** All three counterfactual policies — K subsidy, Pigouvian surcharge on $h(a) \cdot E$, age-25 mandate — *reduce* social welfare at $E = \$17k$. RBP is a partial fix; closing the residual wedge requires direct intervention beyond the actuarial price signal.





Premium and Constructed Per-Tank Expected Loss Across Age Bins

Backup view: per-tank EL = $\hat{h} \times \hat{L} / n_{\text{tanks}}$, where $\hat{L}$ is a control-state donor curve (no TX cost data exi

